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A comprehensive survey of research towards AI-enabled unmanned aerial systems in pre-, active-, and post-wildfire management

Sayed Pedram Haeri Boroujeni · Abolfazl Razi · +7 authors · K. Vamvoudakis

Environmental Science, Engineering · Information Fusion · 4 January 2024

TLDR This **survey** aims to bridge the gaps by offering a systematic **review** of the recent state-of-the-art technologies, highlighting the advancements of UAV systems and AI models from pre-fire, through the active-fire stage, to post-fire management. Expand

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Survey of Wildfire Effects on the Peak Flow Characteristics

F. J. Pirani · P. Coulibaly · Environmental Science · Water resources management · 25 April 2025

Peak flow characteristics, including magnitude, duration, frequency, time to peak, and timing shift, undergo significant changes following **wildfire** disturbances. This study synthesizes findings from… Expand

4 | Springer Nature | Save | Cite

Wildfire Impacts on Security of Electric Power Systems: A Survey of Risk Identification and Mitigation Approaches

Rakibuzzaman Shah · Nima Amjady · Nusrat Subah Binte Shakhawat · Muhammad Usman Aslam · Md. Sazal Miah

Environmental Science, Engineering · IEEE Access · 2024

The increase of extremely hot days globally has led to more frequent **wildfires** and frequent power outages directly and indirectly caused by the **wildfires**. This has sought an urgency to investigate… Expand

6 | IEEE | Save | Cite

Wildfire smoke and health impacts: a narrative review

L. Rizzo · Maria Cândida V. Rizzo · Environmental Science, Medicine · Jornal de Pediatria · 1 December 2024

TLDR Preventive measures against **wildfire** spread must be reinforced, considering the increasing trends of global warming and extreme weather events, including the improvement of early warning systems, improvement of health care facilities and household preparedness and promotion of risk communication campaigns. Expand

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Emerging Trends in Wildfire Detection Through the Lens of Computer Vision and Wildfire Emission Quantification: A Comprehensive Survey

Khushal Das · Jiju Poovvancheri · Sergio Flesca · Claudia Roberta Calidonna · Dong Chen

Environmental Science, Computer Science · IEEE Access · 2026

TLDR This **survey** provides a comprehensive **review** of state-of-the-art computer vision techniques for both **wildfire** detection and emissions quantification, covering 27 detection studies and 16 emissions studies published between 2021 and 2024. Expand

IEEE | Save | Cite

AI for Wildfire Management: From Prediction to Detection, Simulation, and Impact Analysis—Bridging Lab Metrics and Real-World Validation

Nicolas Caron · Hassan N. Noura · Lise Nakache · C. Guyeux · Benjamin Aynes

Environmental Science, Computer Science · Applied Informatics · 1 October 2025

TLDR By bridging the gap between laboratory-oriented models and real-world validation, this

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Climate Change and Wildfire Resilience: Assessing Rural Community Wildfire Protection Plans in the Pacific Northwest

Brent S. Steel E. Wolters Environmental Science, Sociology · Public Organization Review ·

1 December 2025

As **wildfires** intensify with climate change, rural communities in Oregon and Washington increasingly rely on Community **Wildfire** Protection Plans to improve resilience. This study analyzes **survey** data… Expand

Springer Nature Save Cite

AI-based wildfire and smoke detection

Yining Wei Lei Bao Yinlong Liu Jing Sun Jianping Cai Han Yang

Environmental Science, Computer Science · 2025 International Conference on Computer… · 7 November 2025

TLDR This **survey** provides a comprehensive and systematic **review** of the state-of-the-art in AI-based **wildfire** and smoke detection, and analyzes the technological advancements across four primary data acquisition modalities: ground-based camera networks, Unmanned Aerial Vehicles, satellite surveillance, and integrated sensor nodes. Expand

IEEE Save Cite

Wildfire displacement in the United States: a qualitative synthesis of the social consequences

J. A. Gonzalez Dana R. Fisher Environmental Science, Sociology ·

Environmental Research: Climate · 25 February 2026

Wildfire-driven displacement is an urgent and underexamined dimension of social vulnerability in the United States. This study synthesizes findings from a qualitative systematic **review** of the core… Expand

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Survey on Coverage Path Planning with Unmanned Aerial Vehicles

T. Cabreira L. Brisolara Paulo R. Ferreira Jr. Engineering, Environmental Science · Drones ·

4 January 2019

TLDR This paper aims to explore and analyze the existing studies in the literature related to the different approaches employed in coverage path planning problems, especially those using UAVs, and addresses simple geometric flight patterns and more complex grid-based solutions considering full and partial information about the area of interest. Expand

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